



Outbreak of canine distemper and coinfections in a maned wolf (*Chrysocyon brachyurus*) and in three giant anteaters (*Myrmecophaga tridactyla*)

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Abstract

Canine distemper outbreak and coinfections in three giant anteaters and in a maned wolf has been described. Three giant anteaters developed respiratory and digestive clinical signs after the introduction of a maned wolf to a Wildlife Rehabilitation Center. The maned wolf and two anteaters died, and one anteater was euthanized. Post mortem and histopathologic exams revealed lesions associated with numerous intraepithelial inclusion bodies, mainly in the respiratory and digestive systems. Infection by distemper virus was confirmed in all animals by RT-PCR and gene sequencing, which revealed the Europe 1/ South America 1 strain, closely related to the strain from *Canis familiaris*. In addition to distemper, the animals had other comorbidities, such as toxoplasmosis and salmonellosis in the maned wolf and cutaneous candidiasis in an anteater. Considering the chronology of clinical manifestation in both species and the viral characterization, it is possible that the maned wolf was the source of infection to the anteaters. This study demonstrates the importance of implementing biosecurity measures in enclosures that house animals of different species, highlighting the importance of quarantine before introduction of new animals into the same environment.

Keywords Canine distemper virus · Xenarthra · Wild canids · Salmonellosis

Introduction

The giant anteater (*Myrmecophaga tridactyla*) is a placental mammal that inhabits all Brazilian biomes. It is the largest representative of the Myrmecophagidae family, order Pilosa, reaching 45 kg [1]. This species is considered “vulnerable” according to the International Union for Conservation of Nature (IUCN) Red List of Threatened Species, with their population in active decline. Among the main threats to the species, the destruction of natural environments, trampling, burning, hunting, and the occurrence of diseases stand out [2, 3]. The maned wolf (*Chrysocyon brachyurus*) is the largest Brazilian canid, preferring open grass fields, such as the Brazilian savannah. This species is classified as “almost threatened” based on the IUCN Red List of Threatened Species [4]. The main threats to the species include habitat loss due to the expansion of agricultural frontiers and the occurrence of infectious diseases [5]. In the case of endangered species, an understanding of the effect of diseases on the

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health status of these animals is important for the formulation of conservationist policies and for the maintenance of viable populations in the natural environment.

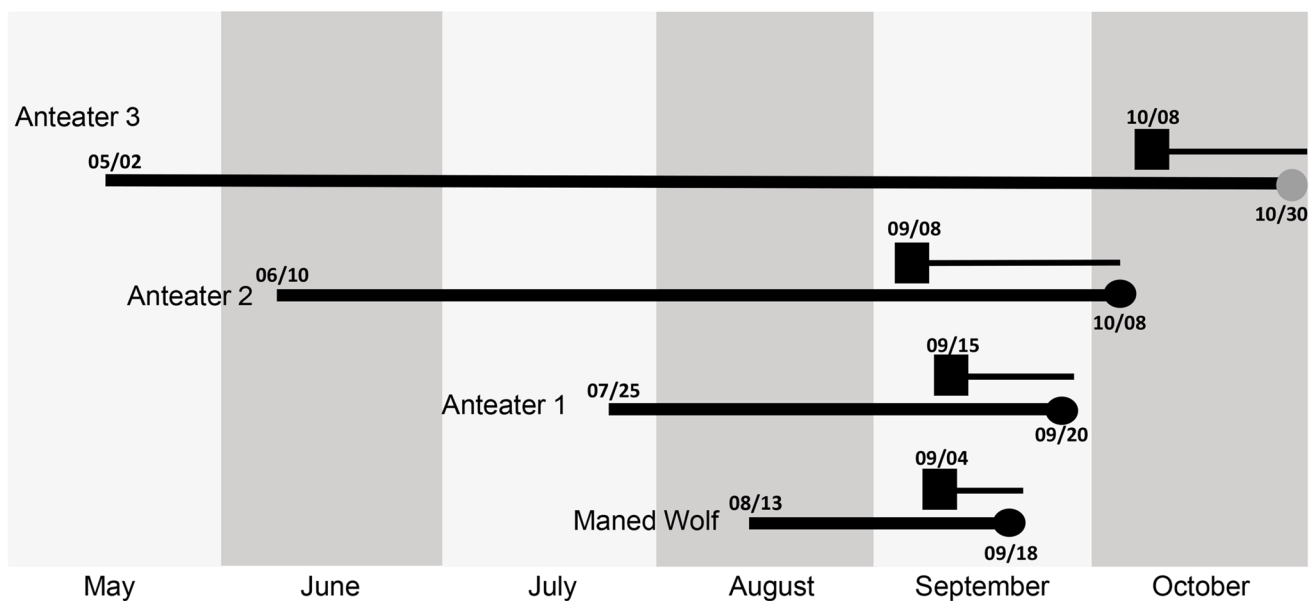
Canine distemper virus (CDV; family Paramyxoviridae, genus *Morbillivirus*) causes high mortality and morbidity in domestic dogs and has been reported to affect wild species in the Canidae, Procyonidae, Mustelidae, Hyaenidae, Ursidae, Viverridae, Felidae, Tayassuidae, Cercopithecidae [6], and Myrmecophagidae families [7]. In 1994, an outbreak of CDV caused the death of 30% of the individuals in an estimated population of 3,000 African lions (*Panthera leo*) [8]. In addition, lethal infections have been described in non-carnivore species such as peccaries and non-human primates demonstrating the remarkable ability of the pathogen to cross species barriers. Secondary infections, including salmonellosis and *Toxoplasma gondii* infections, have also been reported in animals affected by distemper virus [6].

The objective of this study was to describe an outbreak of CDV with coinfections in a young maned wolf and three young giant anteaters, in which CDV was presumed acquired from a maned wolf.

Materials and methods

Animals and samples

The outbreak occurred at a Brazilian wildlife animal rehabilitation center, located in Belo Horizonte, Minas Gerais, Brazil, that housed a 12-week-old male maned wolf and three 6- to 11-week-old giant anteaters (two females [anteaters 2 and 4] and one male [anteater 3]). A timeline of all the events is presented in Fig. 1. All the animals were rescued near roads or close to urbanized areas and were maintained for 20 days (maned wolf) and for 2 to 3 months before death (giant anteaters). The four animals were kept in the same room and were released daily into a common solarium area shared by turtles (*Trachemys* spp.) and tortoises (*Chelonoidis* spp.). The giant anteaters were fed a diet consisting of goat milk, egg yolks, sour cream, and cat food. The maned wolf was fed a diet consisting of fruits, a powdered calcium supplement, slaughtered mice, and dog food. All the animals presented with respiratory and digestive clinical signs. Blood was collected from maned wolf and anteater 2 for complete blood count (CBC) and biochemical tests. Feces were also collected from all the animals for parasitological tests using the Willis-Molloy simple flotation method [9]. A clinical treatment was instituted after a clinical examination, and with progression of the disease, three animals (maned wolf



Legend: — Admitted period at the conservation center ■ Onset of clinical signs ● Euthanasia ● Death

Fig. 1 Time line including the admission period (month and day) of the four animals at the conservation center, disease onset, and disease evolution time

and anteaters 2 and 3) died naturally, and one (anteater 4) was euthanized (Fig. 1). Few hours after death, a complete necropsy was performed in each animal, and tissue sections from multiple organs were collected. Samples for ancillary tests (reverse transcription polymerase chain reaction [RT-PCR] and bacteriology) were also collected and stored appropriately.

Histopathology

Tissue sections were fixed in neutral-buffered 10% formalin for 48 h, trimmed, embedded in paraffin, sectioned at 5- μ m thickness, routinely processed, stained with hematoxylin and eosin (HE) [10], and examined under a standard light microscope. To score the inclusion bodies (IBs) in the tissues, the IB indices were defined as the number of inclusions observed in five consecutive high-power fields (HPF), without overlap, on prioritizing the epithelial tissue of all four animals. Based on this analysis, the intensity of IBs in each tissue was classified as mild (up to 5 IB in 5 HPF), moderate (6 to 15 IB in 5 HPF), and marked (15 or more IB in 5 HPF) [11].

Immunohistochemistry (IHC) and special stains

Samples from the brain, lungs, and liver of maned wolf were subjected to IHC and were tested to detect *T. gondii* and/or *N. caninum* antigens, according to methods described in previous studies [12]. Positive controls consisted of brain samples of dogs naturally infected by *T. gondii* and *N. caninum* (further confirmed by IHC). For the negative control, the primary antibodies were replaced with PBS. Periodic acid-Schiff (PAS) staining was performed on skin samples from anteater 4 [10].

RT-PCR and sequencing for CDV

Samples from the intestine, lungs, liver, kidneys, lymph nodes, and spleen from each animal were individually macerated and diluted tenfold in phosphate buffered saline (PBS). The suspension was vortexed and centrifuged at 4,000 $\times$ g for 10 min at 4 °C, and the supernatant was processed for ribonucleic acid (RNA) extraction using TRIzol (Invitrogen, Thermo Fisher Scientific, Waltham, MA, USA), according to the manufacturer's instructions. Total RNA was subjected to semi-nested RT-PCR to amplify part of the conserved CDV nucleoprotein gene (NP) [13]. To identify the genetic lineages and characterize circulating strains of Brazilian CDV in wild animals, primers that amplify the short region of the fusion (F) gene that encodes the signal peptide of the F protein (Fsp) were used. The RT-PCR products of a 681 bp fragment encompassing the Fsp-coding region (405 bp) were amplified using SuperScript One-Step RT-PCR kit reagents

(Invitrogen) as described in previous studies [14]. Products considered positive were purified with the DNA Clean & Concentrator Kit (Zymo Research, Irvine, CA, USA) and were quantified using Qubit Fluorometric Quantitation (Invitrogen). Bidirectional sequencing was performed to verify the specificity of PCR and to confirm the agent (BigDye Terminator v.3.1, Cycle Sequencing Kit, Applied Biosystems, Foster City, CA, USA; ABI 3500 sequencer, Applied Biosystems). The quality of the sequences was determined using Phred Electropherogram Quality Analysis software (<http://asparagin.cenargen.embrapa.br/phph>), and nucleotide sequence data were analyzed and assembled using SeqMan Software 7.1.0 (DNASTAR, Inc. Madison, WI, USA).

Sequences were aligned with references obtained from the GenBank database using MAFFT v7.471 [15] and the globalpair method to adjust the direction option. Phylogeny was estimated by maximum likelihood using IQ-TREE [16], where the best nucleotide substitution model selected by the program was HKY + F + I [17]. Branch support was estimated using 1,000 bootstrap replicates. The tree image was generated using the ITOL online tool [18], and the final tree was rooted at the midpoint root.

Salmonella sp. isolation and antimicrobial susceptibility and similarity by enterobacterial repetitive intergenic consensus (ERIC)-PCR

Samples from intestinal content, lung, liver, mesenteric lymph nodes, and spleen from each animal, were submitted to isolation of *Salmonella* spp. as previously described [19]. In addition, cloacal swab samples from cohabiting reptiles ($n = 10$) were also tested for the presence of *Salmonella* spp. The identification of *Salmonella* genus was performed by PCR [20], and the identification of the specific serovar was performed according to the antigenic characterization reported at the Brazilian National Reference Laboratory of Enterobacteriaceae of the Oswaldo Cruz Foundation (FIOCRUZ, Brazilian Ministry of Health) [21]. The isolated strains were subjected to antimicrobial susceptibility testing using the disc diffusion method according to the Clinical and Laboratory Standards Institute (CLSI) manual, VET01-A4 [22]. The following antimicrobials were tested: chloramphenicol, trimethoprim/sulfamethoxazole, ceftiofur, ceftriaxone, amoxicillin/clavulanic acid, ampicillin, tetracycline, nalidixic acid, enrofloxacin, and ciprofloxacin. Additionally, to evaluate the genetic similarity among *Salmonella* spp. isolates, one strain from each animal was fingerprinted by ERIC-PCR using the primers ERIC-1 and ERIC-2 as previously described [19, 23]. The genetic patterns were analyzed using Bionumerics 7.6 (Applied Maths, Sin-Martens-Latem, Belgium). *Salmonella* Typhimurium ATCC 14,028 was used as the control.

Results

History and clinical signs

Common clinical signs in the four animals included dehydration, diarrhea, abdominal distension, weight loss, hyporexia, and prostration. All clinical signs, described individually, are shown in Supplemental Table 1. Diarrhea, hematochezia, and dyspnea were among the most severe clinical signs. Ataxia was observed in anteaters 2 and 4. Clinical progression occurred in 10 days in the maned wolf and 2 to 3 weeks in the anteaters. Empirical treatment was carried out in all four animals. Parasitological tests revealed eggs of the Ancylostomatidae family in the maned wolf, and this animal was dewormed with oral fenbendazole (50 mg/kg). CBC and serum biochemistry performed in maned wolf and anteater 2 revealed normochromic anemia, moderate thrombocytopenia, and neutrophilia. In addition, alkaline phosphatase levels and hypoproteinemia were increased in the serum (Supplemental Table 2).

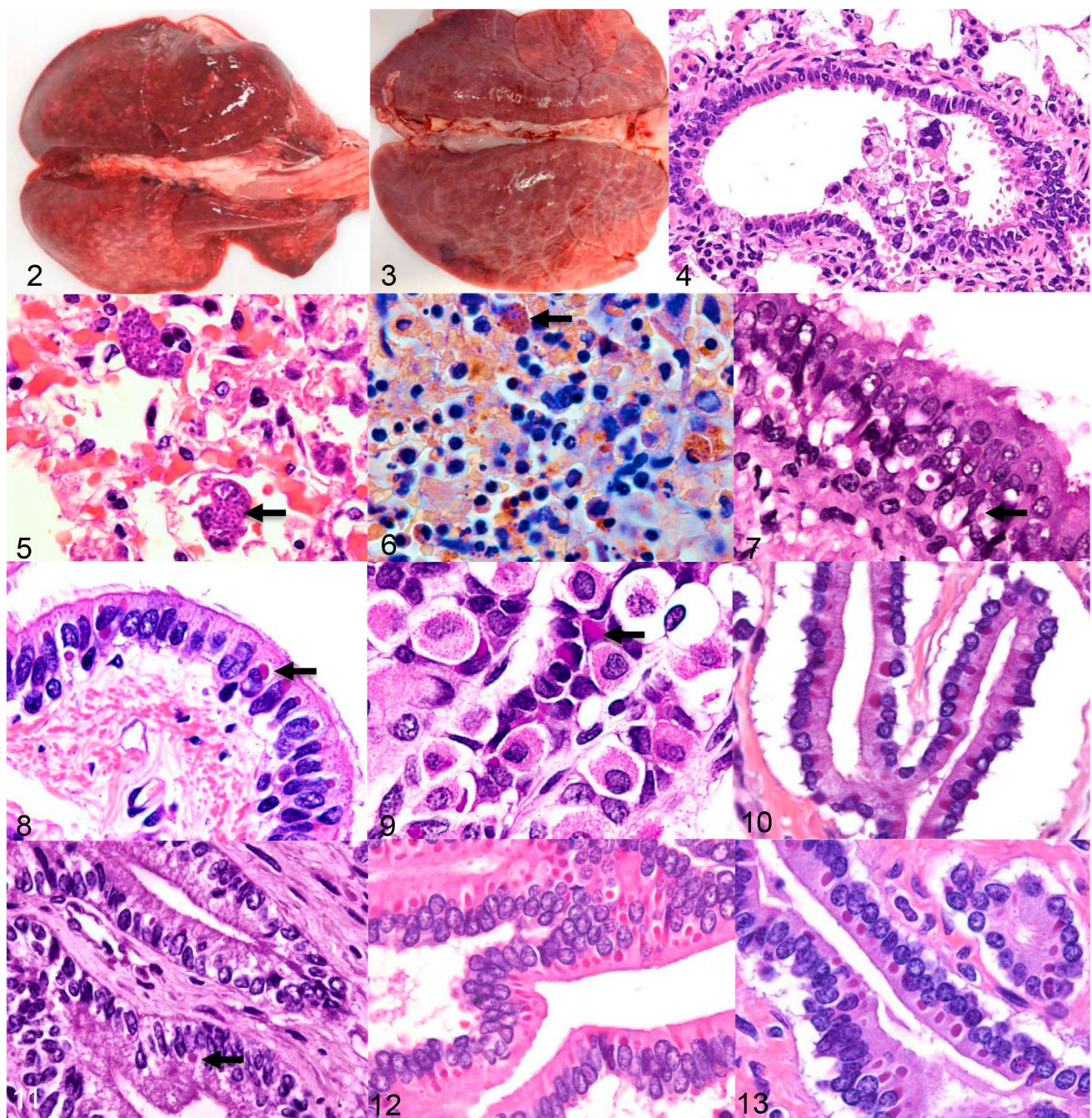
Gross pathology

Grossly, lesions associated with anemia (pale oral and ocular mucous membranes and decreased blood viscosity) were evident among the four animals. The superficial, mediastinal, and mesenteric lymph nodes of the maned wolf were moderately enlarged, with whitish parenchyma and no corticomedullary distinction. In the lungs of all the animals, there was diffuse hyperemia, marked edema, and lobular dark-red consolidation, mostly in the dorsal portion of the cranial and caudal lobes (Figs. 2 and 3). In anteater 3, areas of hemorrhage varying from millimetric to 1.0 cm were observed. There was a mild to moderate hydrothorax (maned wolf and anteaters 3 and 4) and hydroperitoneum (anteaters 3 and 4). In all animals, the liver was pale red to yellowish, with multiple millimetric white foci in the parenchyma in maned wolf and an accentuated lobular pattern in anteaters 2 and 3. The small intestinal serosa of all the animals was hyperemic. In the small intestinal mucosa in maned wolf 1, there were multiple petechial/ecchymosis hemorrhages, together with numerous small helminths, identified as *Ancylostoma caninum*, and some larger helminths, identified as *Toxocara* spp. The intestinal lumen of the four animals was partially filled with fluid contents. In maned wolf and anteaters 3 and 4, there was also a bloody content accumulation. In the brain of all the animals, leptomenigeal hyperemia was observed, and multiple petechiae were also found in anteater 3. In the skin of the ventral abdomen and medial side of the limbs in anteaters 3 and 4, there were multiple 0.5-cm brown crusts. A full list of gross findings is available in Supplemental Table 3.

Fig. 2-13 2 Maned wolf. Lung. Non-collapsed, severely hyperemic, and edematous with multifocal to coalescing dark-red areas. 3 Anteater 3. Lung. Diffusely and severely hyperemic with lobular consolidation especially seen on dorsal lobar regions, characterizing an interstitial pneumonia. 4 Anteater 3. Lung. Detachment of bronchiolar epithelial cells and syncytia formation (center). Hematoxylin and eosin (HE). 5 Maned wolf. Lung. Alveolar septum and lumen with two bradyzoites (arrow), surrounded by fibrin exudate. Hematoxylin and eosin (HE). 6 Maned wolf. Mediastinal lymph node. Focally extensive area of necrosis with tachyzoites and bradyzoites strongly immunoreactive (arrows). Immunohistochemistry for *Toxoplasma gondii*. 7 Intracytoplasmic and eosinophilic inclusion bodies in multiple organs, compatible with infection by the canine distemper virus (*Morbillivirus*). Hematoxylin and eosin. Anteater 2. Trachea. The intracytoplasmic and eosinophilic inclusion bodies in the epithelia are indicated by an arrow. 8 Intracytoplasmic and eosinophilic inclusion bodies in multiple organs, compatible with infection by the canine distemper virus (*Morbillivirus*). Hematoxylin and eosin. Anteater 3. Lung, bronchiolar epithelium. The intracytoplasmic and eosinophilic inclusion bodies in the epithelia are indicated by an arrow. 9 Intracytoplasmic and eosinophilic inclusion bodies in multiple organs, compatible with infection by the canine distemper virus (*Morbillivirus*). Hematoxylin and eosin. Anteater 3. Stomach, oxyntic glands. The intracytoplasmic and eosinophilic inclusion bodies in the epithelia are indicated by an arrow. 10 Intracytoplasmic and eosinophilic inclusion bodies in multiple organs, compatible with infection by the canine distemper virus (*Morbillivirus*). Hematoxylin and eosin. Anteater 3. Duodenum, epithelium of submucosal glands. 11 Intracytoplasmic and eosinophilic inclusion bodies in multiple organs, compatible with infection by the canine distemper virus (*Morbillivirus*). Hematoxylin and eosin. Anteater 2. Uterus, endometrial glands. The intracytoplasmic and eosinophilic inclusion bodies in the epithelia are indicated by an arrow. 12 Intracytoplasmic and eosinophilic inclusion bodies in multiple organs, compatible with infection by the canine distemper virus (*Morbillivirus*). Hematoxylin and eosin. Anteater 3. Liver, bile duct epithelium. 13 Intracytoplasmic and eosinophilic inclusion bodies in multiple organs, compatible with infection by the canine distemper virus (*Morbillivirus*). Hematoxylin and eosin. Anteater 3. Pancreas, pancreatic duct epithelium (exocrine pancreas)

Histopathology

Histopathological findings revealed IBs in a variety of tissue sections, mainly in the epithelia and intracytoplasmic, in all the animals (Table 1). Respiratory and digestive epithelia were the main site of intracytoplasmic IBs, and several cells were swollen or had hydropic degeneration. The type, intensity, and distribution of lesions sorted by animal are shown in Table 2. In the lungs of all the animals, there was an interstitial pneumonia (bronchiointerstitial in maned wolf and anteater 3), characterized by lymphoplasmacytic infiltrate in the alveolar septa. Multinucleated syncytial cells with fibrin exudation were present in the bronchi and bronchioles in maned wolf and anteater 3 (Fig. 4). Moreover, there was diffuse alveolar edema with multiple foci of hemorrhage and areas of fibrinous exudate in all the animals. In the liver, there was centrilobular to midzonal hydropic degeneration (maned wolf and anteater 2 and 4), centrilobular to midzonal coagulative necrosis (anteater 2), and fibrinonecrotic and neutrophilic random



hepatitis (anteaters 3 and 4). In the spleen and lymph nodes of all four animals, there was a lymphoid depletion. Lympholysis was present in the spleen in maned wolf and anteater 2, while extramedullary hematopoiesis was observed in the spleen of all the animals. In maned wolf, there were additional lesions in the parenchyma of the lung, liver, and spleen, characterized by multiple fibrinonecrotic areas surrounded by numerous neutrophils and macrophages. Similar lesions were found in the mediastinal, tracheobronchial, and mesenteric lymph nodes, replacing most of the lymphoid parenchyma.

Inside the necrotic foci, there were multiple protozoal cysts with bradyzoites, measuring approximately 30 μm in diameter (Fig. 5), and free or intracellular tachyzoites, measuring 3 μm in diameter. IHC results were positive for *T. gondii* (Fig. 6) and negative for *N. caninum* in the lung, liver, and brain samples in maned wolf. Neurological lesions were found in two animals. In maned wolf, there were few areas in the mesencephalon with microgliosis, sometimes associated with necrosis and protozoal cysts. In anteater 3, there was lymphohistiocytic meningitis with fibrinoid vasculitis, along with lymphocytic and neutrophilic choroiditis. No

Table 1 Organs or tissues with intracytoplasmic viral inclusion bodies and quantitative evaluation

Organs	Maned wolf	Anteater 2	Anteater 3	Anteater 4
Lungs (respiratory epithelium of trachea, bronchi and bronchioles)	+	+++	+++	+++
Liver (biliary duct epithelium)	+++	++	+++	+++
Small and large intestine (crypt enterocytes)	*	++	+++	+++
Duodenum (submucosal glands)	*	*	++	+++
Pancreas (pancreatic ductal epithelium)	++	*	+++	++
Nasal turbinates (respiratory epithelium of nasal mucosa)	-	-	+++	-
Liver (hepatocytes or Kupffer cells)	*	*	++	*
Urinary bladder (transitional epithelium)	*	*	*	++
Skin (epidermis)	*	*	+	+
Spleen (lymphocytes and histiocytes cytoplasm)	*	*	+	*
Stomach (chief and parietal cells)	*	*	+	*
Kidneys (transitional epithelium of renal pelvis)	*	*	*	+
Tongue (mucosal epithelium)	*	+	*	+
Uterus (endometrial epithelium and endometrial glands)	-	+	-	*
Conjunctiva (mucosal epithelium)	*	*	+	*

The organs on this list are ordered according to the presence of viral inclusion bodies: +, mild; ++, moderate; +++, severe; *not observed on this animal; -, does not apply/tissue sample not available

neurological lesions compatible with CDV were observed in these animals. Cutaneous lesions of hyperkeratosis associated with numerous pseudohyphae and blastoconidia (PAS-stained) compatible with *Candida* spp. were found in anteater 4. Intracytoplasmic IBs of CDV in several different tissues are shown in Figs. 7–13.

RT-PCR and sequencing for CDV

The CDV RT-PCR was positive in four animals on testing tissues from the following different organs: spleen (maned wolf), liver and lungs (anteaters 2 and 3), and kidneys and mesenteric lymph nodes (anteater 4).

Table 2 Major histopathological findings considering distribution and severity of lesions in affected animals

Histopathological findings	Maned wolf		Anteater 2		Anteater 3		Anteater 4	
	G	Di	G	Di	G	Di	G	Di
Lymphoplasmacytic interstitial pneumonia	+	M	+	M	++	D	+	M
Bronchointerstitial pneumonia	+	M	*	*	+	M	*	*
Fibrinonecrotic pneumonia	++	M	*	*	*	*	*	*
Pulmonary hemorrhage	+	M	+	M	+++	M	+	M
Pulmonary edema	++	M	++	M	++	M	++	M
Chronic neutrophilic and lymphohistioplasmacytic myocarditis	*	*	*	*	+++	M	*	*
Centrolobular hepatic necrosis	*	*	+++	D	*	*	*	*
Random fibrinonecrotic hepatitis	++	M	*	*	+	M	+	M
Hepatic degeneration	*	*	+	D	++	D	++	D
Extramedullary hematopoiesis	*	*	+++	D	+++	D	+++	D
Lymphoid depletion	++	D	+++	D	+++	D	++	D
Splenic lympholysis	*	*	++	M	*	*	*	*
Fibrinonecrotic lymphadenitis	+++	M	*	*	*	*	*	*
Lymphohistiocytic encephalic vasculitis	*	*	*	*	+++	D	*	*
Lymphohistiocytic choroiditis	*	*	*	*	++	D	*	*
Adrenal hemorrhage	*	*	++	M	++	M	++	M
Orthokeratotic hyperkeratosis	*	*	*	*	+	M	++	M
Epidermitis	*	*	*	*	*	*	++	M

G, grade; mild, +; moderate, ++; severe, +++; Di, distribution; M, multifocal; D, diffuse; *absence of histologic lesions

Table 3 GenBank accession numbers of the sequenced samples from maned wolf and all three anteaters

Sample ID	Host species	GenBank accession number
LG1.maned wolf/1/UFGM/BR/2020	<i>Chrysocyon brachyurus</i>	MZ437800
TB2.giant anteaters/2/UFGM/BR/2020	<i>Myrmecophaga tridactyla</i>	MZ437797
TB3.giant anteaters/3/UFGM/BR/2020	<i>Myrmecophaga tridactyla</i>	MZ437798
TB4.giant anteaters/4/UGMG/BR/2020	<i>Myrmecophaga tridactyla</i>	MZ437799

The phylogenetic analysis showed that sequences from this study (Table 3, Fig. 14, in red) were clustered with other isolates from Brazil within the same clade and genotype Europe 1/South America 1. The sequences in turn were aligned within a subcluster, diverging from the other Brazilian samples.

Although all Brazilian samples were part of the same cluster and genotype (Europe 1/ South America 1), dissimilarities were still observed among the samples, as demonstrated by a p-distance analysis. The isolates obtained

in this study showed similarities ranging from 97.2 to 99.8%. Sequences from anteater 3 were the least similar (97.2–98.4%) when compared to the others. Sequences from maned wolf shared high genetic similarity with sequences from anteaters 2 (99.8%), 3 (99.5%), and 4 (97.9%), which were derived from a *Myrmecophaga tridactyla* host. Maned wolf shared 96.3% to 98.4% similarity with other Brazilian sequences derived from dogs (*Canis lupus familiaris*) (KY057352 and KY057353) as well as from *Cerdocyon*

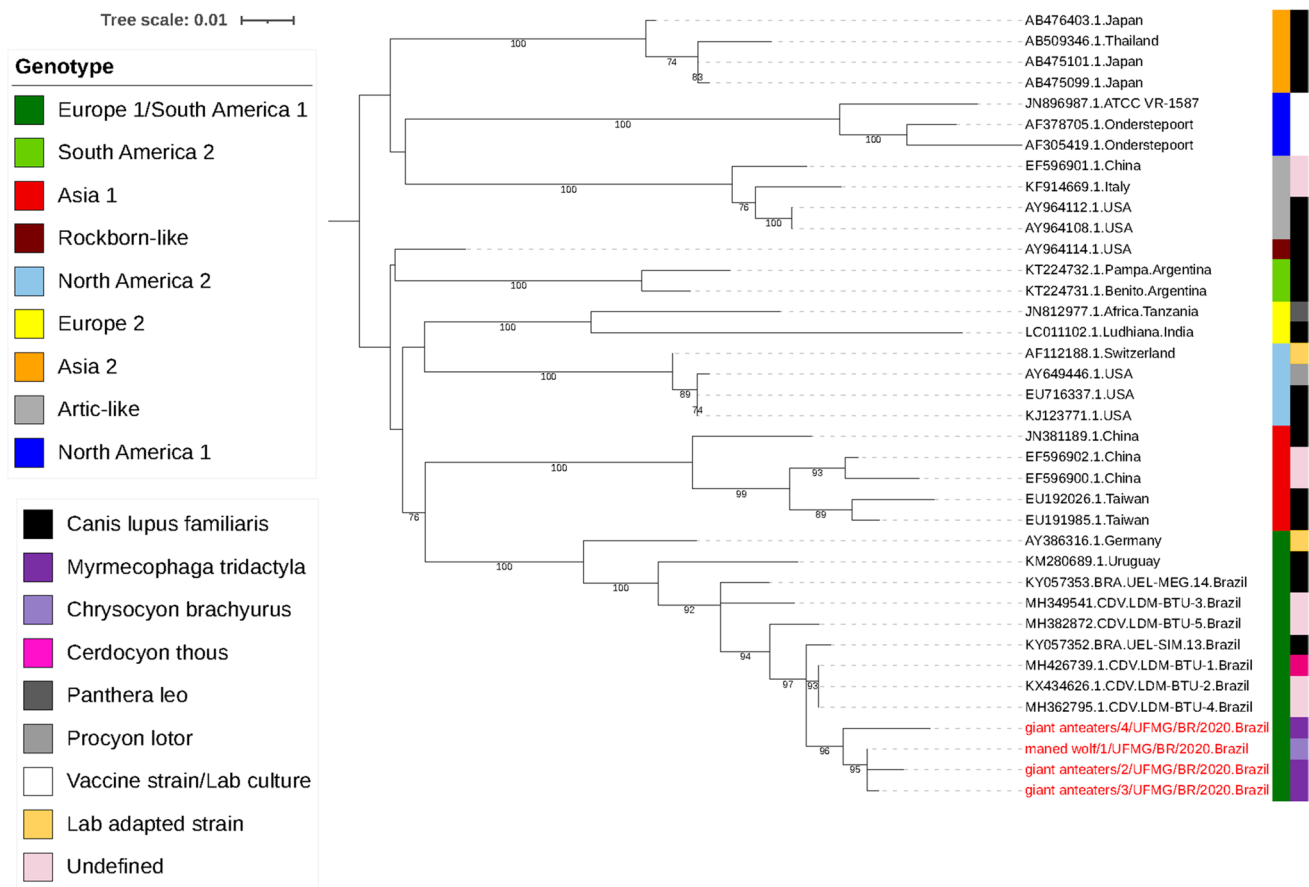


Fig. 14 Nucleotide maximum likelihood phylogenetic analysis of the fusion (F) gene that encodes the signal peptide of the F protein (Fsp) of canine distemper virus strains. Samples written in red indicate the sequenced samples of this study. The first and second colored stripes

from left to right represent the genotype and host from which the isolates were obtained, respectively. Bootstrap values were determined in 1000 replications; only bootstrap values $\geq 70\%$ are shown

thous (98.6%) species (MH426739.1). In turn, *Myrmecophaga tridactyla*-derived sequences from this study (anteaters 2, 3, and 4) shared 95.1% to 98.1% similarity with other Brazilian sequences derived from *Canis lupus familiaris* species and 97.5–98.4% similarity with those derived from *Cerdocyon thous* (MH426739.1) species.

Compared with sequences from other countries, those in this study were more similar to German (AY386316.1) and Uruguayan (KM2806689.1) sequences (91.4–94%), both classified as the Europe 1/South America 2 genotype. High genetic similarities were also observed with a sequence from Rockborn-like and North America 2 genotypes, with values ranging from 89.1 to 90.7% and 85.9 to 87.7%, respectively. Similarity of 85.4–87.7% was observed with the South America 2 genotype. The most divergent sequences compared to those of the Brazilian samples were vaccine strains (Onderstepoort AF378705.1 and AF305419.1) from the North America 1 genotype, with similarity values ranging from 79.9 to 81.9%. Lower similarities (81.5–83.8%) with genotype Arctic-like were also observed.

Salmonella sp. isolation and antimicrobial susceptibility and similarity by ERIC-PCR

Salmonella sp. was isolated from the intestinal content and from several other organs in the maned wolf and three giant anteaters (anteaters 2, 3, and 4). Serotyping revealed four *Salmonella* serovars: *S. Typhimurium* was isolated in maned wolf (liver, spleen, mesenteric lymph nodes, and intestinal content), *S. Panama* was isolated in anteater 2 (liver, spleen, mesenteric lymph nodes, and intestinal content), *S. Agona* was isolated in anteater 3 (intestinal content, liver, and lung),

and *S. Saintpaul* was isolated in anteater 4 (feces, liver, and spleen) (Supplemental Table 4). Three *Salmonella* serovars were also isolated from four of the 10 cloacal swab samples from the cohabiting reptiles (Supplemental Table 5): *S. Seftenberg*, *S. Typhimurium*, and *S. Panama*. All *Salmonella* isolates obtained in the present study were fully susceptible to the antimicrobials tested. The ERIC-PCR also showed a low degree of genetic similarity between the *Salmonella* spp. isolates (Fig. 15) from all of the affected animals and from the healthy cohabiting reptiles.

Discussion

Due to the chronology of the events, it is believed that the young maned wolf housed into the rehabilitation center was responsible for introducing the disease into the enclosure, since the affected three anteaters did not present any clinical signs indicative of the disease prior to the arrival of the maned wolf. This hypothesis is supported by the high similarity of genetic material revealed by the phylogenetic analysis of the four animals in this study. Epidemiological studies based on serological and/or molecular methods (RT-PCR) for the detection of CDV in a population of wild canids that were found dead or among live animals without clinical signs have shown that this virus is frequent in these populations [24, 25]. The presence of unvaccinated stray dogs in peri-urban areas and anthropized zones may be an important source of viral infection in wild animals [24].

The similarity between the gross and microscopic findings found in these animals and wild canids, including the red fox (*Vulpes vulpes*) and the Italian wolf (*Canis lupus*

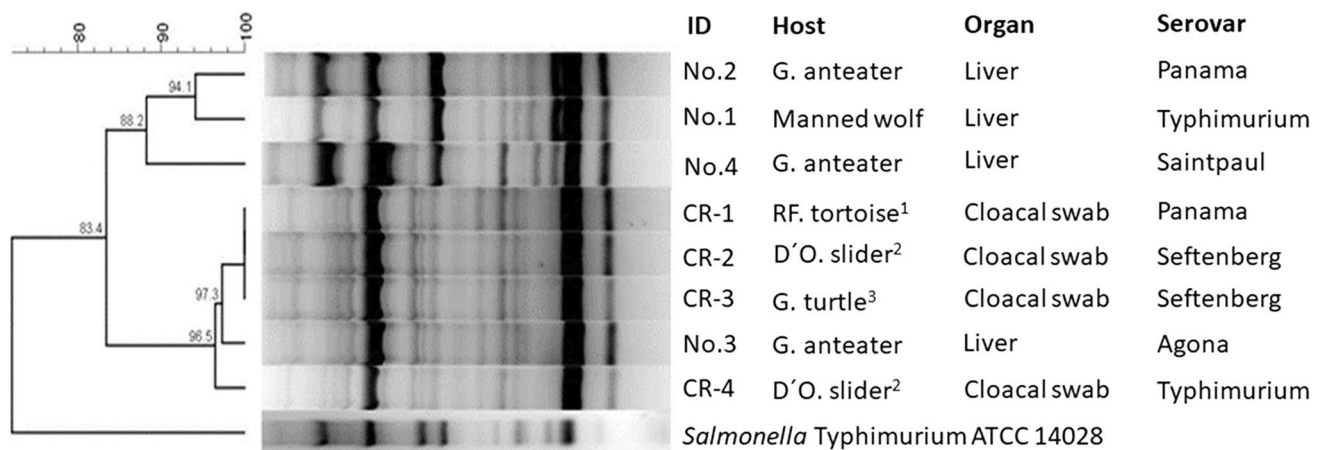


Fig. 15 Dendrogram by enterobacterial repetitive intergenic consensus polymerase chain reaction (ERIC-PCR) of eight *Salmonella* spp. isolates from the affected animals and cohabiting reptiles during the outbreak of canine distemper virus. Clustering analysis was based on the Dice similarity coefficient and unweighted pair group method by arithmetic mean (UPGMA) algorithms using Bionumerics 7.6

(Applied Maths, Belgium). From the right to the left: identification (ID), species (host), source of isolation (organ), and serovar of *Salmonella* spp. (serovar). Legend: CR, cohabiting reptiles, ¹red-footed tortoise (*Chelonoidis carbonarius*), ²D'Orbigny's slider (*Trachemys dorbignyi*), and ³Geoffroy's side-necked turtle (*Phrynops geoffroanus*)

italicus) [25], as well as in other mammals, such as Asian palm civets (*Paradoxurus hermaphroditus*) [26] and the giant panda (*Ailuropoda melanoleuca*) [27], should also be highlighted. The lesions associated with viral infection in the animals in the present study were mainly characterized by interstitial or bronchointerstitial pneumonia; IB formation in the epithelium of multiple tissues, especially in the respiratory and digestive tract, together with in the lymphoid tissues, sometimes accompanied by lympholysis and lymphoid depletion [26]. Respiratory and intestinal lesions were also described in a giant anteater from the Brazilian Midwest, in addition to cutaneous lesions (nasal plane and paw pad hyperkeratosis) [28]. The neurological form of canine distemper has not yet been described in giant anteaters, but its occurrence in a collared anteater (*Tamandua tetradactyla*) has recently been documented [7]. The finding of meningoencephalitis in one of the anteaters was not associated with the viral infection, since demyelination and IBs were not found within neurons and glial cells at the site of injury, and fibrinoid vasculitis and neutrophilic choroiditis indicated systemic bacterial infection [29].

Distemper lesions in the digestive tract of the animals in this study, characterized by hydropic degeneration and intracytoplasmic eosinophilic IBs in enterocytes, biliary ducts, and gastric parietal cells, were also previously described in a giant anteater [28] and in Asian palm civets [26]. However, viral inclusions in the intestinal submucosal glands, pancreatic ducts, and uterine glands have only been described here. Similarly, in other reports, affected animals presented with diarrhea, which was associated with the digestive form of distemper [26, 28, 30], along with intestinal endoparasites (helminths). Furthermore, no intestinal lesions were found that could be attributed to the enteric form of salmonellosis.

Blood tests from the maned wolf and an anteater (anteater 2) revealed mild lymphopenia, which may be associated with lymphoid depletion due to viral infection, since distemper virus tends to replicate in lymphocytes causing lympholysis. In dogs, this finding frequently predisposes to secondary bacterial and/or parasitic infections [31], some of which were diagnosed in the animals in this study. Coinfections were evident in three out of the four cases, and the lesions in the maned wolf were consistent with those in a case of acute systemic toxoplasmosis [32], characterized by encephalitis, and necrotizing pneumonia, hepatitis, lymphadenitis, and splenitis, with multiple intralesional bradyzoites and tachyzoites. Positive immunostaining for *T. gondii* confirmed coinfection caused by the protozoa, which was possibly related to the decrease in immunity caused by the previous viral infection. Moreover, the presence of stray cats near the rehabilitation center may be an epidemiological indicator of a possible source of infection [31, 33].

The association between bacteriological findings and random fibrinous and necrotizing hepatitis indicated that the

infection by *Salmonella* Agona and *Salmonella* Saintpaul in the anteaters in anteaters 2 and 3, respectively, were possibly responsible for the lesion. Nevertheless, all four animals in the present study were positive for *Salmonella* sp. and all isolated strains were fully susceptible to the antimicrobials tested. This coinfection is not surprising, as CDV is known to cause severe immunosuppression, which can facilitate coinfections, including salmonellosis [34]. *Salmonella* spp. were also isolated from four of the 10 tested reptiles that were housed in the same area. Because of this result, it was initially hypothesized that these infections could be due to the same *Salmonella* sp. clone, which might have been transmitted among the affected animals, probably from a reptile. In contrast to this hypothesis, different serovars were identified in each affected animal. In addition, the ERIC-PCR also showed a low degree of genetic similarity between the *Salmonella* spp. isolates (Fig. 15) from all of the affected animals and from the healthy cohabiting reptiles (CR1 to 4). Together, these results refute the hypothesis of a single *Salmonella* sp. source for these coinfections and did not confirm reptiles as a source of infection.

Cutaneous infection by *Candida* spp. in anteater 4 may have also been caused by viral immunosuppression [31]. Hepatic degeneration and centrilobular necrosis in the first anteater may be explained by tissue hypoxia due to severe pulmonary impairment caused by interstitial pneumonia [35].

In the present study, all the affected animals were rescued by environmental police in areas of peri-urban highways (anteaters) or near burning Cerrado savannah areas (maned wolf). Anteater mothers are commonly run over, and cubs are found alone. This indicates that reduction and fragmentation of environments may have an influence on the occurrence of fatal accidents in wild animals, generally followed by the development of infectious diseases in the rescued animals in rehabilitation centers like the present one. CDV positive serological results from wild canids in the same region were obtained in a previous study [24].

Previous phylogenetic studies performed in Brazil showed that the Europe 1/South America-1 genotype was mainly present in samples derived from domestic dogs [36]. Sequences from the present study were clustered in the same clade of this genotype. However, these sequences were aligned within a subcluster, demonstrating a high similarity among them and low similarity with other Brazilian samples.

Overcrowding or lack of space to separates different species of animals in the rehabilitation centers is a predisposing factor for the occurrence of infectious disease outbreaks among wild animals. Providing conditions to ensure the application of biosafety measures is essential for the adequate management of these species. Here, the associations between clinical signs, epidemiology, gross and

histopathologic findings, and ancillary diagnostic test results were important for the clinical and epidemiological understanding as well as for the characterization of the agents involved in this outbreak.

Conclusions

The present work demonstrates the importance of implementing biosecurity measures in enclosures that house wild animals of different species, thereby highlighting the importance of quarantine for animals that are introduced into such environments. Considering the chronology of clinical manifestation in both species and viral characterization, it is possible that the maned wolf was the source of infection to the anteaters. The numerous intracytoplasmic IBs in different tissues suggest that the virus has a high capacity for replication in anteaters, but further studies are required to explain the absence of distemper-related neurological lesions in these cases, as well as the viremic phases, viral spread patterns, and the role of these animals in the distemper cycle. Thus, CDV infection and coinfections should be included in the list of differential diagnoses in wild mammals that present with apathy, diarrhea, and/or respiratory problems; further, additional tests and clinical-epidemiological data are required to improve the diagnostic approach in these cases.

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Declarations

Conflict of interest The authors declare no competing interests.

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